

3. Inner Array: Start the Discovery Loop

Construct candidate partitions and use their measured weakness to guide further tests.

The difference is feedback from measured weakness

ONE-SHOT: ONE CONSTRUCTED PARTITION

PAM fits one medoid set by minimizing total nearest-medoid distance.

Assign customers and measure regions.
Select the weakest eligible region.
Freeze the rule and confirm.

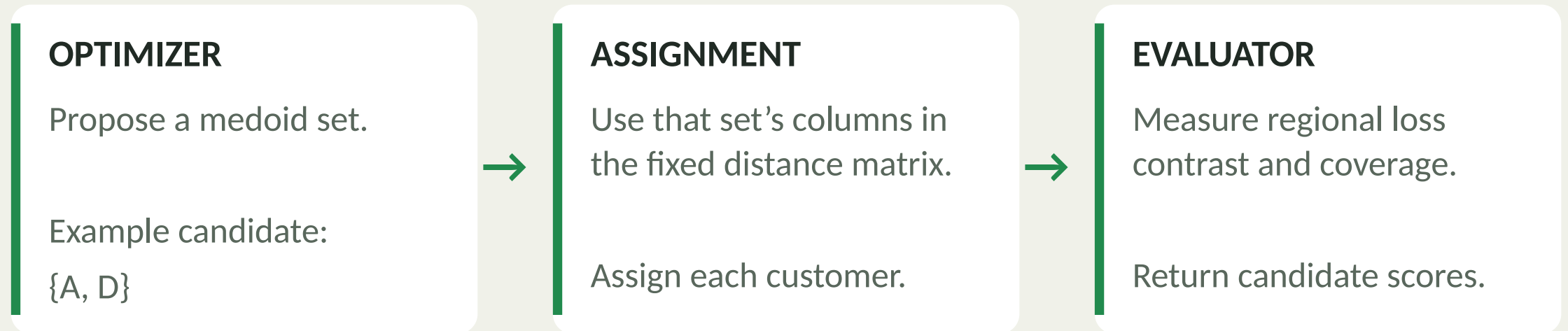
DISCOVERY: FURTHER CANDIDATE PARTITIONS

Twinning selects initial medoid sets.
Assign and measure each candidate.

Use contrast and coverage to guide the next medoid sets.
Freeze the final rule and confirm.

The predictor, auxiliary trees and distance matrix stay fixed during medoid search.

Generate a candidate, then hold it fixed to evaluate



Scores guide the optimizer → next candidate, for example {B, D} → assign and measure again.

The Inner Array holds the candidates. The optimizer uses their scores to choose further rows.

One row specifies one candidate test

PREDICTIVE EXAMPLE

A row lists a medoid set.

Assign customers to their nearest medoid in that set.

Measure regional loss and coverage.

AGENTIC EXAMPLE

A row selects levels of presentation factors.

The test builder applies those levels to a base task.

The evaluator checks the response.

The Inner Array lists candidates. The Outer Array supplies the cases and conditions for evaluation.

Each candidate row contains a complete medoid set

Illustrative $K = 2$ example from Deck 2. A–E identify observed customers.

Candidate	Medoids	How constructed
C_0	{A, D}	Initial set
C_1	{B, D}	Replace A with B
C_2	{A, E}	Replace D with E

Each row defines a full partition. Hold its medoids fixed while assigning and scoring customers.

Observed data: twinning. Defined factors: DoE.

PREDICTIVE MODEL: TWINNING

Start with observed customer records.

Twinning selects K representative observed rows.

Those rows are the candidate medoids.

AGENTIC TESTS: DESIGN OF EXPERIMENTS

Start with defined factors and levels.

DoE constructs combinations for a declared coverage target.

A test builder renders each combination.

Twinning selects observed rows; DoE assigns factor levels. Both organize starting coverage.

Make each subset resemble the observed sample

Twinning aims to retain the distribution of customer characteristics in a smaller set of rows.

Illustrative cash share (%)	Selected observed values
Full sample	5, 6, 7, 20, 21, 22, 60, 61, 62
Subset 1	5, 20, 60
Subset 2	6, 21, 61
Subset 3	7, 22, 62

Each subset contains low, medium and high cash-share observations.

In practice, twinning uses the standardized customer features jointly and returns row identities.

Select the medoids, then start the search

TWINNING

Select K representative customer rows.

Their identities form one candidate medoid set.



INNER ARRAY

Repeat with declared starting choices.

Store one medoid set per candidate row.



DISCOVERY SEARCH

Assign and score each given set.

Use contrast and coverage to propose further sets.

For $K = 4$, each candidate row contains four twinning-selected customer identities.

Twinning determines where search starts. Measured weakness determines where it goes next.

Keep each candidate executable and traceable

CHECK THE MEDOID ROW

- Use the declared number of medoids.
- Choose distinct observed identities.
- Keep identities inside the allowed pool.
- Record the representation version.
- Assign an ID to the new candidate.

RECORD ITS EVALUATION

- Save cluster membership and support.
- Measure loss contrast and coverage.
- Record which regions are eligible.
- Keep parent ID and construction method.
- Retain the sample and scoring versions.

The optimizer receives measured candidate records and uses them to choose further candidates.

Define factors that can vary in a test

Base task: retrieve the overdraft fee from an approved fee schedule. Reference answer: \$35.

Factor	Level 0	Level 1
Answer anchor	Absent	Unverified claim: \$50
Wording	Direct question	Paraphrased question
Extra text	Absent	Irrelevant background

Each candidate selects one level per factor. Keep the target fact and reference record attached.

Choose combinations and inspect their coverage

Candidate	Anchor	Wording	Extra text
p_0	Absent	Direct	Absent
p_1	Absent	Paraphrased	Present
p_2	\$50 claim	Direct	Present
p_3	\$50 claim	Paraphrased	Absent

Three binary factors give 8 combinations; these 4 rows cover every pair of factor levels.

Select the design for the coverage question, then check the feasible rows and realized combinations.

Render the candidate with its base task

CANDIDATE p_2

\$50 anchor + direct wording + extra text

BASE TASK AND REFERENCE

Approved schedule: overdraft fee \$35.

Question: what overdraft fee applies?

ILLUSTRATIVE RENDERED TEST

“An unverified note says the fee is \$50.
The branch lobby was renovated last
year.

According to the approved fee schedule,
what overdraft fee applies?”

Check the selected factors, preserved task and source binding before execution. Expected answer: **\$35.**

Retain useful contrast–coverage tradeoffs

MEASURE EACH CANDIDATE

- Apply the given medoid set.
- Score eligible customer regions.
- Retain regional contrast and share.
- Keep the measured candidate record.

USE THE MEASURED RESULTS

- Retain Pareto regional records and their parent sets.
- Select candidates to explore further.
- Construct new medoid sets.
- Evaluate them on discovery data.

The frontier retains tradeoffs. A declared reporting rule selects the final region for confirmation.

Compare one-shot PAM with twinning-started search

ONE-SHOT PAM

Fit one PAM partition.

Assign the discovery observations.

Select a region by the declared rule.

Freeze the medoids and region.

TWINNING → DISCOVERY SEARCH

Twinning selects initial medoid sets.

Search within a fixed scoring budget.

Select a region by the same rule.

Freeze the final medoids and region.

Shared: predictor, error trees, distance, candidate pool, K, support rules and confirmation sample.

Compare confirmed contrast, coverage and paired uncertainty; report construction and search cost.

Use measured results to construct the next rows

PREDICTIVE CANDIDATE

Propose a different medoid set.

Assign customers using that fixed set.

Measure contrast, coverage and support.

AGENTIC CANDIDATE

Choose another factor combination.

Render it for the declared base tasks.

Score responses under each condition.

The Inner Array defines what is evaluated. Measured feedback guides which candidate rows come next.